Lab4: Conditional VAE for Video Prediction

311351009 劉承霈

I. Introduction

In this lab, I worked on making video prediction using the VAE method. I used a sequence of images showing poses in the video, along with one starting image, to create a new video that predicts what might happen. In addition, I used some clever techniques like Teacher forcing strategy, KL Annealing, and Reparameterization tricks to improve the model's performance.

II. Implementation Details

1. How do you write your training protocol?

- **Data Preparation**: First, the dimensions of the tensors are reshaped from (B, seq, C, H, W) to (seq, B, C, H, W). This step sets up the data to be processed frame by frame in the upcoming stages.
- **Feature Extraction**: Human features are extracted from the initial frame's image. These features are then saved as a reference template for the upcoming frames.
- **Teacher Forcing**: The model decides whether to use Teacher Forcing. If chosen, the model combines previous frame features and its output to predict human features for the current frame.
- **Posterior Prediction**: The predicted human features and label features are passed through the Gaussian Predictor to obtain a latent variable z, along with mean and variance.
- **Decoder Fusion**: The model combines the predicted human features, label features, and latent variable z to compute the decoding parameters.
- **Image Generation**: The Generator generates a new image output based on the calculated decoding parameters.

- **Loss Calculation and KL Annealing**: The mean squared error and KL divergence for the current frame are computed and accumulated. The current KL divergence is weighted by beta determined by the KL annealing process.
- **Backpropagation, Optimization and Loss Return**: Backpropagation is performed to compute gradients. The optimizer then updates the model's parameters based on these gradients. The computed total loss is returned as the outcome of the training step.

2. How do you implement reparameterization tricks?

- **Calculate Standard Deviation**: Compute the standard deviation by exponentiating half of the logarithm of the variance, converting it to the original scale.
- **Generate Random Noise**: Sample random noise from a standard normal distribution (N(0, 1)) with the same shape as std. This introduces randomness for exploring diverse latent values.
- **Reparameterization**: Compute the latent variable z by adding the product of noise and std to the mean. This maintains differentiability, allowing gradients to flow through stochastic operations.
- **Return Latent Variable**: Return the computed latent variable z as the outcome of reparameterization.

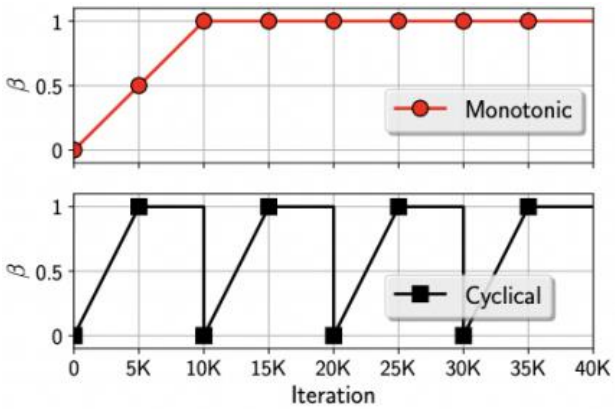
3. How do you set your teacher forcing strategy?

- If adapt_TeacherForcing is true, the model calculates the predicted human features for the current frame as a weighted combination of the human features from the previous frame and the output's human features.
- If adapt_TeacherForcing is false, the predicted human features are solely based on the output's human features from the previous frame. In this scenario, the model relies entirely on its own predictions

without referencing the ground truth.

- As training progresses and epochs increase, the teacher forcing ratio decreases, leading the model to rely less on the ground truth and encouraging it to generate outputs based on its own predictions.

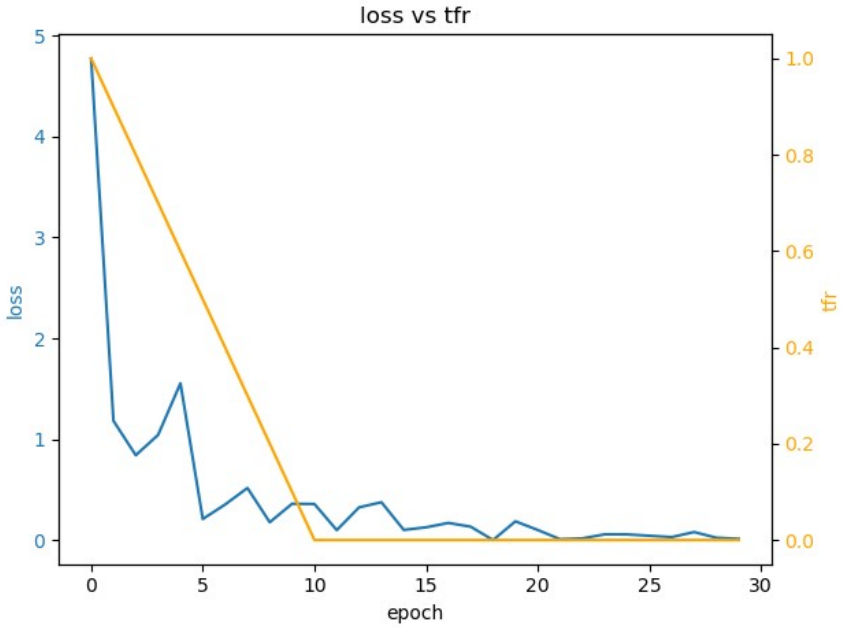
4. How do you set your kl annealing ratio?



- The KL annealing ratio is set based on the annealing type. If the annealing type is "Cyclical," the ratio is determined by the current position within the cycle and is calculated using the frange_cycle_linear() function. On the other hand, if the annealing type is "Monotonic," the ratio is increased by 0.1 in each update until it reaches 1.0. If none of these conditions are met, the KL annealing ratio is set to 1.0 directly.

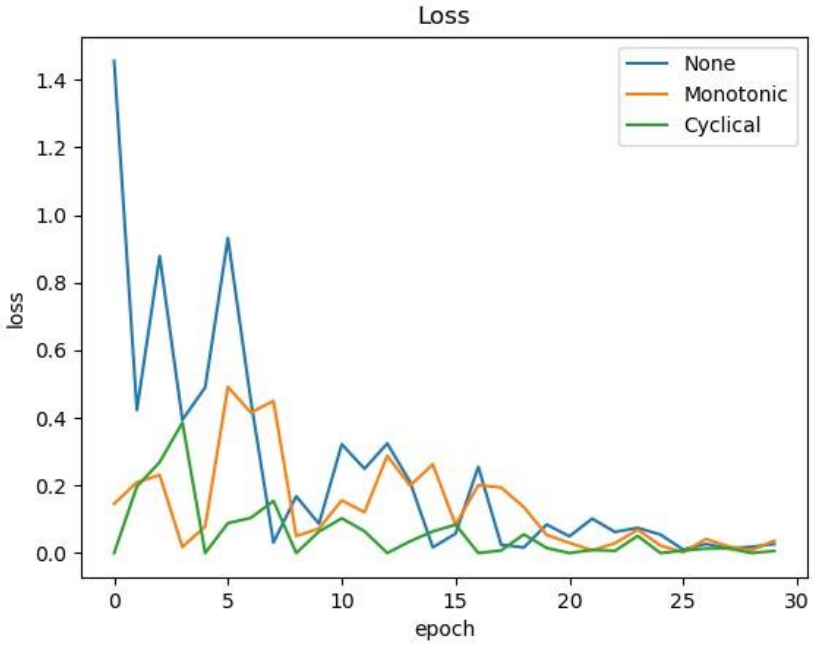
III. Analysis & Discussion

1. Plot Teacher forcing ratio. Analysis & compare with the loss curve



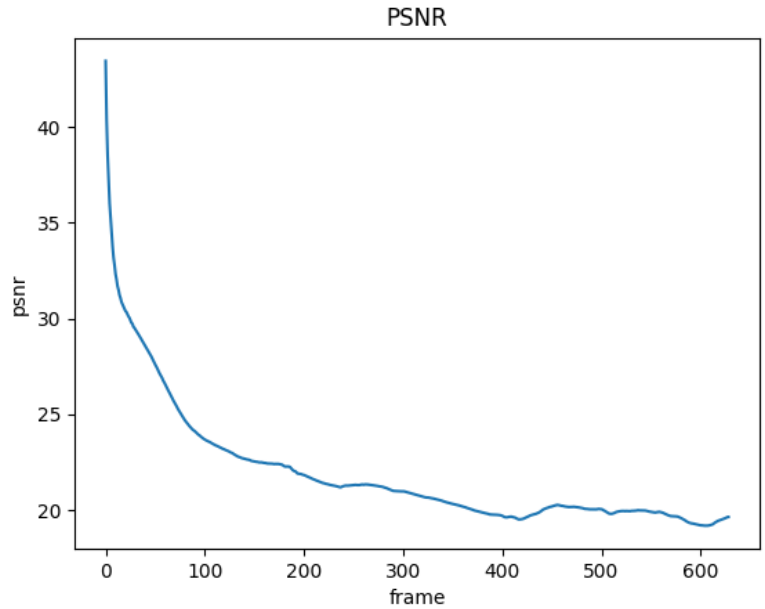
- The teacher forcing ratio drops 0.1 per epoch in the first 10 epochs, so the loss overall decreases. But occasionally the loss increases, that's because we're not using teacher forcing strategy then. After 10 epochs, the curve tends to be stable since the teacher forcing strategy isn't used.

2. Plot the loss curve while training with different settings. Analyze the difference between them.



- **Without KL annealing**: The loss curve shows erratic behavior or slow convergence. The KL divergence term's impact from the start can lead to unstable training and higher chances of getting stuck in poor local minima.
- **With KL Annealing (Monotonic)**: The loss curve demonstrates more stable and smooth convergence. Initially, the KL term's weight is low, focusing primarily on data reconstruction. As training progresses, the KL term's weight gradually increases, leading to a smoother transition in the loss curve and better utilization of the latent space.
- **With KL Annealing (Cyclical)**: The loss curve might exhibit fluctuations in the KL term's weight. This can result in periods of intense focus on data reconstruction and exploration of latent space alternately. The curve could show a cyclical pattern with relatively faster convergence during phases of higher KL weight.
- Therefore, incorporating KL annealing techniques, whether monotonic or cyclical, tends to lead to more controlled training, potentially faster convergence, and improved exploration of the latent space compared to training without annealing.

3. Plot the PSNR-per frame diagram in validation dataset



4. Derivate conditional VAE formula

與原本的 VAE 一樣

$P(x|c;\theta) = \int P(x|z,c;\theta)P(z)\,dz$ is intratable

採用 EM algorithm，並從 chain rule 可得：

$\log P(x|c;\theta) = \log P(x,z|c;\theta) - \log P(z|x,c;\theta)$

兩邊同乘 $q(z)$，並對 z 積分：

$\int q(z)\log P(x|c;\theta)\,dz$

$= \int q(z)\log P(x,z|c;\theta)\,dz - \int q(z)\log q(z)\,dz$

$+ \int q(z)\log q(z)\,dz - \int q(z)\log P(z|x,c;\theta)\,dz$

$\Rightarrow \log P(x|c;\theta) = \mathcal{L}(x,c,q,\theta) + KL(q(z)\,\|\,P(z|x,c;\theta))$

其中 $\begin{cases} \mathcal{L}(x,c,q,\theta) = \int q(z)\log P(x,z|c;\theta)\,dz - \int q(z)\log q(z)\,dz \\ KL(q(z)\,\|\,P(z|x,c;\theta)) = \int q(z)\log \dfrac{q(z)}{P(z|x,c;\theta)}\,dz \end{cases}$

又 $\because$ KL divergence ≥ 0

$\therefore \log P(x|c;\theta) \geq \mathcal{L}(x,c,q;\theta)$

且 if and only if $q(z) = P(z|x,c;\theta)$，則上式等號成立

因此 $\mathcal{L}(x,c,q;\theta)$ 為 $P(z|x,c;\theta)$ 的 lower bound

接著將 $q(z|x,c;\theta')$ 代入 $\mathcal{L}(x,c,q;\theta)$：

P.S: $P(x,z|c) = P(z|c)P(x|z,c)$

$\mathcal{L}(x,c,q;\theta) = E_{z\sim q(z|x,c;\theta')}\log P(x|z,c;\theta)$

$+ E_{z\sim q(z|x,c;\theta')}\log P(z|c) - E_{z\sim q(z|x,c;\theta')}\log q(z|x,c;\theta')$

$= E_{z\sim q(z|x,c;\theta')}\log P(x|z,c;\theta) - KL(q(z|x,c;\theta')\,\|\,P(z|c))$ #